

Creation Date 15-Feb-2011

Revision Date 25-Jan-2024

Revision Number 4

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                               |
|---------------------------|-------------------------------|
| Product Description:      | <b>Cadmium, coarse powder</b> |
| Cat No. :                 | <b>78251</b>                  |
| Index No                  | 048-002-00-0                  |
| CAS No                    | 7440-43-9                     |
| EC No                     | 231-152-8                     |
| Molecular Formula         | Cd                            |
| REACH registration number | -                             |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

### 1.3. Details of the supplier of the safety data sheet

|         |                                                                                                                                                                                                              |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Company | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608 |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                |                                |
|----------------|--------------------------------|
| E-mail address | begel.sdsdesk@thermofisher.com |
|----------------|--------------------------------|

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

**CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567**

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

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Acute Inhalation Toxicity - Dusts and Mists  
Germ Cell Mutagenicity  
Carcinogenicity  
Reproductive Toxicity  
Specific target organ toxicity - (repeated exposure)

Category 2 (H330)  
Category 2 (H341)  
Category 1B (H350)  
Category 2 (H361fd)  
Category 1 (H372)

## **Environmental hazards**

Acute aquatic toxicity  
Chronic aquatic toxicity

Category 1 (H400)  
Category 1 (H410)

Full text of Hazard Statements: see section 16

## **2.2. Label elements**



Signal Word

Danger

## **Hazard Statements**

H330 - Fatal if inhaled  
H341 - Suspected of causing genetic defects  
H350 - May cause cancer  
H361fd - Suspected of damaging fertility. Suspected of damaging the unborn child  
H372 - Causes damage to organs through prolonged or repeated exposure  
H410 - Very toxic to aquatic life with long lasting effects

## **Precautionary Statements**

P201 - Obtain special instructions before use  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P310 - Immediately call a POISON CENTER or doctor/physician  
P273 - Avoid release to the environment

## **Additional EU labelling**

Restricted to professional users

## **2.3. Other hazards**

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment

Toxic to terrestrial vertebrates  
Toxicity to Soil Dwelling Organisms  
This product does not contain any known or suspected endocrine disruptors

## **SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

### **3.1. Substances**

| Component | CAS No    | EC No             | Weight % | CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567 |
|-----------|-----------|-------------------|----------|-----------------------------------------------------------------------------------------|
| Cadmium   | 7440-43-9 | EEC No. 231-152-8 | 100      | Acute Tox. 2 (H330)                                                                     |

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|  |  |  |  |                                                                                                                                 |
|--|--|--|--|---------------------------------------------------------------------------------------------------------------------------------|
|  |  |  |  | Muta. 2 (H341)<br>Carc. 1B (H350)<br>Repr. 2 (H361fd)<br>STOT RE 1 (H372)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) |
|--|--|--|--|---------------------------------------------------------------------------------------------------------------------------------|

| Component | Specific concentration limits (SCL's) | M-Factor | Component notes |
|-----------|---------------------------------------|----------|-----------------|
| Cadmium   | -                                     | 10       | -               |

|                           |   |
|---------------------------|---|
| REACH registration number | - |
|---------------------------|---|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Use personal protective equipment as required. Avoid contact with skin, eyes or clothing. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                     |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Notes to Physician | Treat symptomatically. |
|--------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

**Suitable Extinguishing Media**  
approved class D extinguishers. Do not use water or foam.

**Extinguishing media which must not be used for safety reasons**  
Water.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. Do not allow run-off from fire-fighting to enter drains or

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water courses.

## Hazardous Combustion Products

Cadmium oxide.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance.

#### Hygiene Measures

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing. Avoid contact with skin, eyes or clothing. Wash hands before breaks and immediately after handling the product. Keep away from food, drink and animal feeding stuffs.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510      Class 6.1B  
Storage Class (LGK) (Germany)

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

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## Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE** - 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

| Component | The United Kingdom                                                                       | European Union                    | Ireland                                                                                                                                                                                                                                               |
|-----------|------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cadmium   | STEL: 0.075 mg/m <sup>3</sup> 15 min<br>TWA: 0.025 mg/m <sup>3</sup> 8 hr<br>Carc. metal | TWA: 0.001 mg/m <sup>3</sup> (8h) | TWA: 0.001 mg/m <sup>3</sup> 8 hr. inhalable fraction<br>TWA: 0.004 mg/m <sup>3</sup> 8 hr. limit value 0.004 mg/m <sup>3</sup> until 11 July 2027 inhalable fraction<br>STEL: 0.003 mg/m <sup>3</sup> 15 min<br>STEL: 0.012 mg/m <sup>3</sup> 15 min |

## Biological limit values

List source(s):

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                    | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Cadmium<br>7440-43-9 ( 100 ) |                                  |                                     | DNEL = 4µg/m <sup>3</sup>          |                                       |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                    | Fresh water     | Fresh water sediment        | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)      |
|------------------------------|-----------------|-----------------------------|--------------------|------------------------------------|-------------------------|
| Cadmium<br>7440-43-9 ( 100 ) | PNEC = 0.19µg/L | PNEC = 1.8mg/kg sediment dw |                    | PNEC = 20µg/L                      | PNEC = 0.9mg/kg soil dw |

| Component                    | Marine water    | Marine water sediment        | Marine water intermittent | Food chain            | Air |
|------------------------------|-----------------|------------------------------|---------------------------|-----------------------|-----|
| Cadmium<br>7440-43-9 ( 100 ) | PNEC = 1.14µg/L | PNEC = 0.64mg/kg sediment dw |                           | PNEC = 0.16mg/kg food |     |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Nitrile rubber |                                   |                 |             |                       |
| Neoprene       |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

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Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

## Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Particulates filter conforming to EN 143

## Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Particle filtering: EN149:2001

When RPE is used a face piece Fit Test should be conducted

## Environmental exposure controls

Local authorities should be advised if significant spillages cannot be contained. Do not allow material to contaminate ground water system. Prevent product from entering drains.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Physical State                          | Solid Granules           |                                   |
| Appearance                              | Silver                   |                                   |
| Odor                                    | Odorless                 |                                   |
| Odor Threshold                          | No data available        |                                   |
| Melting Point/Range                     | 321 °C / 609.8 °F        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | 765 °C / 1409 °F         | @ 760 mmHg                        |
| Flammability (liquid)                   | Not applicable           | Solid                             |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | No information available | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | No information available |                                   |
| Viscosity                               | Not applicable           | Solid                             |
| Water Solubility                        | No information available |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Vapor Pressure                          | No information available |                                   |
| Density / Specific Gravity              | 8.64 @ 25°C              |                                   |
| Bulk Density                            | No data available        |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Particle characteristics                | No data available        |                                   |

### 9.2. Other information

|                   |                        |
|-------------------|------------------------|
| Molecular Formula | Cd                     |
| Molecular Weight  | 112.40                 |
| Evaporation Rate  | Not applicable - Solid |

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## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under recommended storage conditions. Moisture sensitive. Air sensitive.

### 10.3. Possibility of hazardous reactions

#### Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
None under normal processing.

### 10.4. Conditions to avoid

Incompatible products. Exposure to air or moisture over prolonged periods.

### 10.5. Incompatible materials

Strong oxidizing agents. Strong acids. Sulfur oxides.

### 10.6. Hazardous decomposition products

Cadmium oxide.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

No data available

Inhalation

Category 2

| Component | LD50 Oral                 | LD50 Dermal | LC50 Inhalation                            |
|-----------|---------------------------|-------------|--------------------------------------------|
| Cadmium   | LD50 = 2330 mg/kg ( Rat ) | -           | LC50 = 25 mg/m <sup>3</sup> ( Rat ) 30 min |

#### (b) skin corrosion/irritation;

No data available

#### (c) serious eye damage/irritation;

No data available

#### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

#### (e) germ cell mutagenicity;

Category 2

Possible risk of irreversible effects

#### (f) carcinogenicity;

Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | EU           | UK | Germany | IARC    |
|-----------|--------------|----|---------|---------|
| Cadmium   | Carc Cat. 1B |    | Cat. 1  | Group 1 |

#### (g) reproductive toxicity; Reproductive Effects

Category 2

Possible risk of impaired fertility. May cause harm to the unborn child.

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|                                            |                                              |
|--------------------------------------------|----------------------------------------------|
| (h) STOT-single exposure;                  | No data available                            |
| (i) STOT-repeated exposure;                | Category 1                                   |
| Target Organs                              | Blood, Respiratory system, Kidney, Prostate. |
| (j) aspiration hazard;                     | Not applicable<br>Solid                      |
| Symptoms / effects, both acute and delayed | No information available.                    |

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Water Flea                                         | Freshwater Algae |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------|
| Cadmium   | LC50: 0.0004 - 0.003 mg/L, 96h<br>(Pimephales promelas)<br>LC50: = 0.016 mg/L, 96h<br>(Oryzias latipes)<br>LC50: = 21.1 mg/L, 96h<br>flow-through (Lepomis macrochirus)<br>LC50: = 0.24 mg/L, 96h static<br>(Cyprinus carpio)<br>LC50: = 4.26 mg/L, 96h<br>semi-static (Cyprinus carpio)<br>LC50: = 0.002 mg/L, 96h<br>(Cyprinus carpio)<br>LC50: = 0.006 mg/L, 96h static<br>(Oncorhynchus mykiss)<br>LC50: = 0.003 mg/L, 96h<br>flow-through (Oncorhynchus mykiss) | EC50: = 0.0244 mg/L, 48h Static<br>(Daphnia magna) |                  |

| Component | Microtox | M-Factor |
|-----------|----------|----------|
| Cadmium   |          | 10       |

**12.2. Persistence and degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
**Persistence** May persist.  
**Degradability** Not relevant for inorganic substances.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**12.3. Bioaccumulative potential** Product has a high potential to bioconcentrate

**12.4. Mobility in soil** No information available



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## 12.5. Results of PBT and vPvB assessment

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

### Persistent Organic Pollutant

### Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. Should not be released into the environment.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

#### 14.1. UN number

UN2570

#### 14.2. UN proper shipping name

CADMIUM COMPOUND

#### Technical Shipping Name

Cadmium metal

#### 14.3. Transport hazard class(es)

6.1

#### 14.4. Packing group

I

### ADR

#### 14.1. UN number

UN2570

#### 14.2. UN proper shipping name

CADMIUM COMPOUND

#### Technical Shipping Name

Cadmium metal

#### 14.3. Transport hazard class(es)

6.1

#### 14.4. Packing group

I

### IATA

#### 14.1. UN number

UN2570

#### 14.2. UN proper shipping name

CADMIUM COMPOUND

#### Technical Shipping Name

Cadmium metal

#### 14.3. Transport hazard class(es)

6.1

#### 14.4. Packing group

I

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**14.5. Environmental hazards** Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Cadmium   | 7440-43-9 | 231-152-8 | -      | -   | X     | X    | KE-04397 | X    | -    |

| Component | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Cadmium   | 7440-43-9 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

| Component | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                                                                                        | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)                                             |
|-----------|-----------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Cadmium   | 7440-43-9 | -                                                                   | Use restricted. See item 72.<br>(see link for restriction details)<br>Use restricted. See item 23.<br>(see link for restriction details)<br>Use restricted. See item 28.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - 231-152-8 - Carcinogenic, Article 57a; Specific target organ toxicity after repeated exposure, Article 57(f) - human health |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

#### REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

#### Seveso III Directive (2012/18/EC)

| Component | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Cadmium   | 7440-43-9 | Not applicable                                                                            | Not applicable                                                                           |

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Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

| Component                    | ANNEX I - PART 1<br>List of chemicals subject to<br>export notification procedure<br>(referred to in Article 8)                                          | ANNEX I - PART 2<br>List of chemicals qualifying for<br>PIC notification<br>(referred to in Article 11) | ANNEX I - PART 3<br>List of chemicals subject to the<br>PIC procedure<br>(referred to in Articles 13 and<br>14) |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Cadmium<br>7440-43-9 ( 100 ) | i(1) — industrial chemical for<br>professional use<br>sr — severe restriction<br><br>i(2) — industrial chemical for<br>public<br>sr — severe restriction | i — industrial chemical<br>sr — severe restriction                                                      | -                                                                                                               |

<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32012R0649&qid=1604065742303>.

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                                                            |
|-----------|---------------------------------------|------------------------------------------------------------------------------------|
| Cadmium   | WGK3                                  | Krebserzeugende Stoffe - Class I : 0.05 mg/m <sup>3</sup><br>(Massenkonzentration) |

| Component | France - INRS (Tables of occupational diseases)               |
|-----------|---------------------------------------------------------------|
| Cadmium   | Tableaux des maladies professionnelles (TMP) - RG 61,RG 61bis |

| Component                    | Switzerland - Ordinance on the<br>Reduction of Risk from<br>handling of hazardous<br>substances preparation (SR<br>814.81) | Switzerland - Ordinance on<br>Incentive Taxes on Volatile<br>Organic Compounds (OVOC) | Switzerland - Ordinance of the<br>Rotterdam Convention on the<br>Prior Informed Consent<br>Procedure |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Cadmium<br>7440-43-9 ( 100 ) | Prohibited and Restricted<br>Substances                                                                                    |                                                                                       | Annex I - industrial chemical                                                                        |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H330 - Fatal if inhaled

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H341 - Suspected of causing genetic defects  
H350 - May cause cancer  
H361fd - Suspected of damaging fertility. Suspected of damaging the unborn child  
H372 - Causes damage to organs through prolonged or repeated exposure  
H400 - Very toxic to aquatic life  
H410 - Very toxic to aquatic life with long lasting effects

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

### **Prepared By**

Health, Safety and Environmental Department

### **Creation Date**

15-Feb-2011

### **Revision Date**

25-Jan-2024

### **Revision Summary**

New emergency telephone response service provider.

**This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**